
Homework 5

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MA 564 Section A1

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Problem 1

Problem 2

rigorizing will be messy, but just note that the subspace topology is still just the cofinite topology, so if i have a set that doesnt fully cover it, but i have a full cover, then there are at most finitely many sets required in the cover to cover the rest after that first set, sicne its compliment is finite

Problem 3

Let $\mathcal{O}$ be an open cover of A . Then

$$A = \bigcup_{O \in \mathcal{O}} (O \cap A) = \left(\bigcup_{O \in \mathcal{O}} O \right) \cap A.$$

Since A is compact, it admits a finite subcover $\mathcal{O}'$, and thus

$$A = \left(\bigcup_{O \in \mathcal{O}'} O \right) \cap A = \bigcup_{O \in \mathcal{O}'} (O \cap A).$$

Therefore, any open cover in the subspace topology admits a finite subcover in the subspace topology.
[converse is harder]

Problem 4

Problem 5

Problem 6

Problem 7